

Information Technologies for Industrial Engineers

General Information

- **Course Number:** 255411
- **Course Name:** Information Technologies for Industrial Engineers
- **Semester:** 2569-1
- **Instructor:** Nirand Pisutha-Arnond



What do you think about when you hear the word "Information Technologies"?

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Gemini Answer

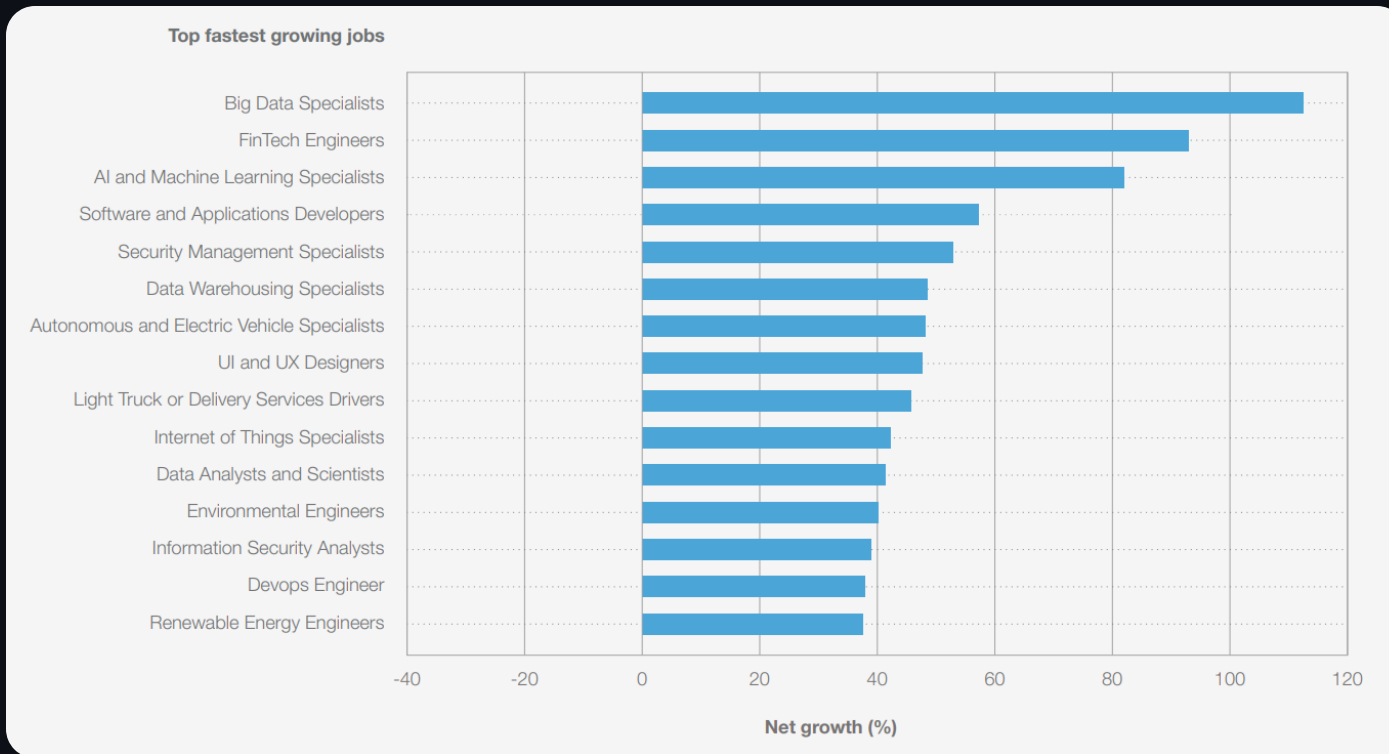
My Experience

- I view myself as an *IT guy*.
 - I teach a course in Computer Engineering department.
 - I work in a software company.
 - I advise many software-related projects in Ministry of Industrial Promotion.
- However, I've never learned Information Technologies directly in school.
- I picked up the knowledge by trying to **build software**.
 - This process introduced me to many different technologies and concepts in Information Technologies.

What I like about being an IT guy

- I feel that I understand how computers work and how to make them work for me.
- I have more tools to help me solve problems.
- It is a basis for the fastest growing jobs according to the World Economic Forum.

The Future of Jobs Report 2025



Source: World Economic Forum, 2025.

Motivation

- Instead of teaching you the theory of Information Technologies, I will teach you how to **build software**.
- This will expose you to many different technologies and concepts in Information Technologies.

After this class you will be able to

1. Develop and deploy *fullstack* application with current software technology
2. Select appropriate technology stack for IT-related problems.

Grading

Evaluation Criteria	Weight
Midterm	20%
Final	20%
Assignments	60%

Assignments will consists of 5-6 small projects.

Course Outline

Fullstack = Frontend + Backend + Deployment

Frontend technology

- HTML
- CSS
- JavaScript
- TypeScript

Backend technology

- Node.js
- Express.js
- Database

Deployment

- Linux (server administration)
- Docker

Extra

If time permits, we *may* cover

- LLM (Large Language Model) and AI (Artificial Intelligence) in the context of web application development.
- Blockchain and Web3 in the context of web application development.

Fullstack Developer Roadmap

Do not panic!

Using AI

- The goal of this course to expose you to the current software technology.
 - Not to make you a professional software engineer.
- Therefore, it is okay to use AI tools to assist your coding.
- However, the more you outsource your tasks to AI, the less you will learn.
So, use it wisely.

Using AI

- Bad way to use AI
 - Use it to code you something you don't understand very well.
- Good way to use AI
 - Use it like a better Google + Stack Overflow.
 - Use it like a code reviewer to help you find mistakes in your code.

Assignment

- You will be asked to explain your code in the VDO along with your submission.
- This encourages you to understand your code and not just copy-paste from AI.

Lets get started

Tools

- **Terminal**
 - Windows - PowerShell 7 + Chocolatey ([Setup Guide](#))
 - MacOS - Zsh + Homebrew
- **Editor: VSCode**
 - Enable: Format on save
 - Extension: Prettier
 - Extension: Auto Rename Tag